

SONAR: A Physics-constrained Neural Representation for X-ray Dark-field CT

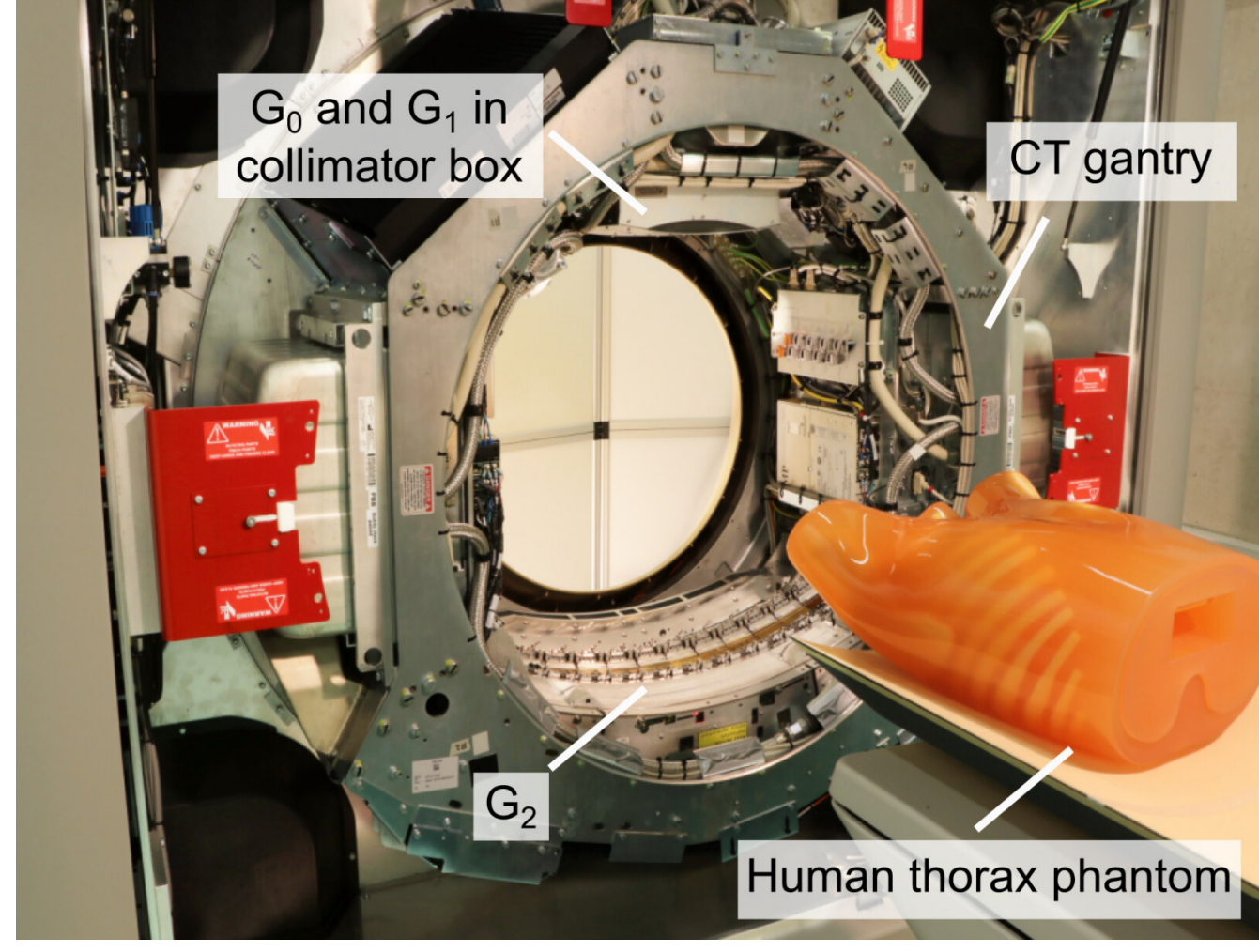
Daniel Frey^{1,2}, Theresa Hiu^{1,2}, Julian McGinnis^{3,4}, Tina Dorosti^{1,2,5}, Johannes B. Thalhammer^{1,2,5,6}, Sebastian Peterhansl^{1,2}, Zijin Huang^{1,2}, Franz Pfeiffer^{1,2,5,6}, Daniel Rueckert^{3,4,7}, Florian Schaff^{1,2}



¹ Chair of Biomedical Physics, TUM School of Natural Sciences, Technical University of Munich (TUM), Germany; ² Munich Institute of Biomedical Engineering, TUM; ³ Chair of AI in Healthcare and Medicine, TUM University Hospital, TUM; ⁴ Munich Center for Machine Learning (MCML), TUM; ⁵ Institute for Diagnostic and Interventional Radiology, TUM University Hospital, TUM; ⁶ TUM Institute for Advanced Study (IAS), TUM; ⁷ Department of Computing, Imperial College London, United Kingdom

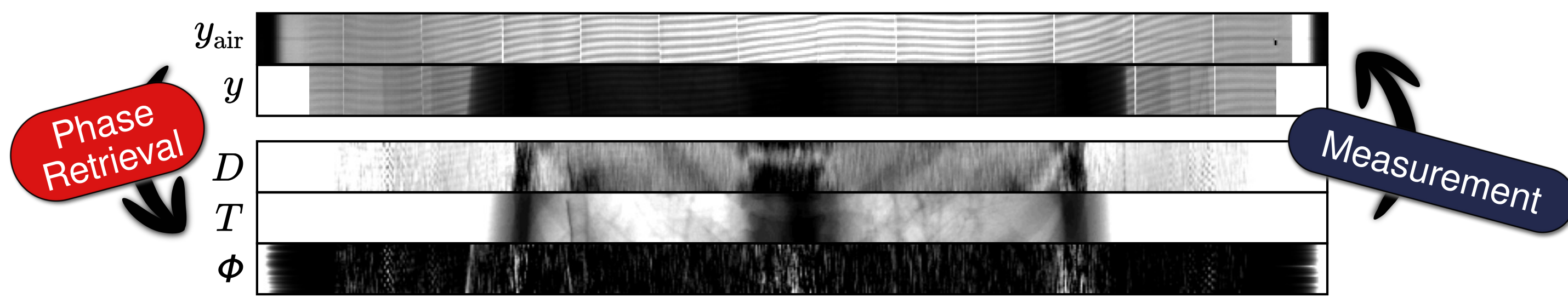
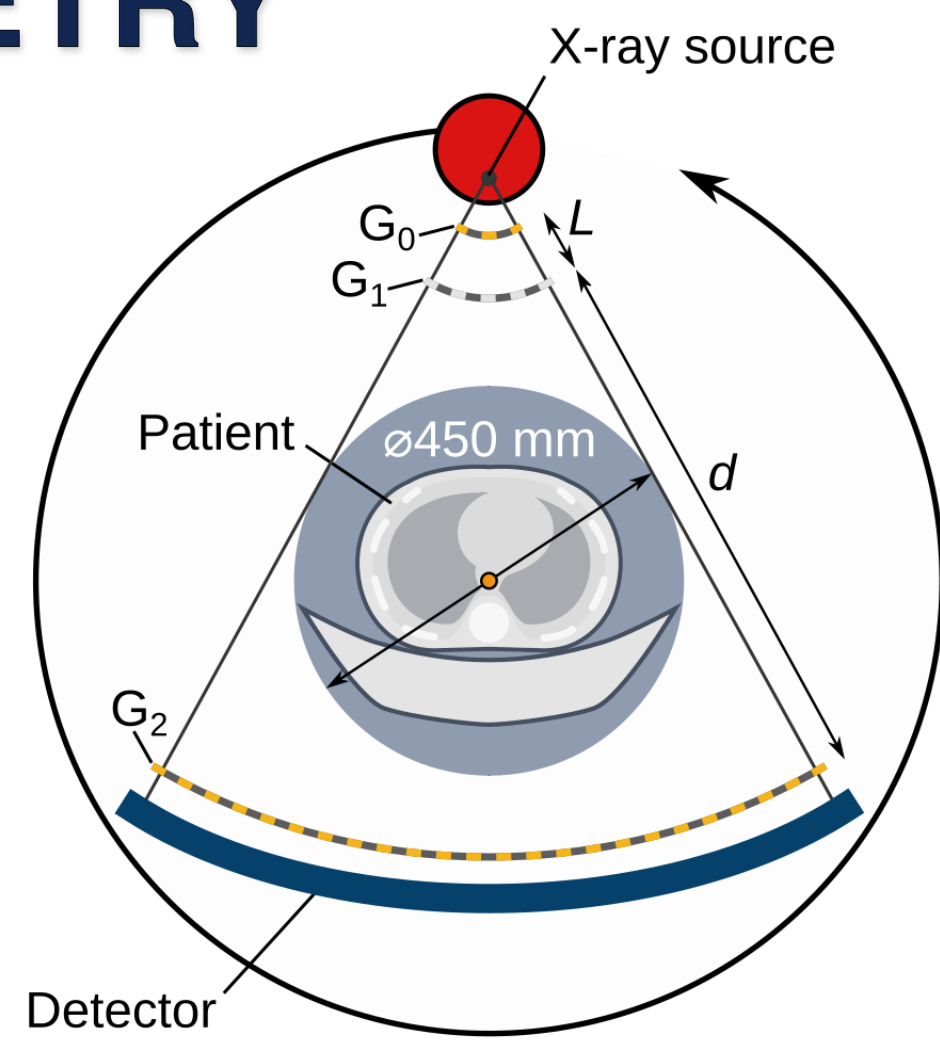
INTRODUCTION

- Human-scale **X-ray dark-field computed tomography (DFCT)** enables functional lung imaging [1]
- Reconstructions suffer from **streak artifacts** due to limited phase sampling in continuously acquired scans [2, 3]
- Implicit neural representations (INRs)** encode discretized images as inherently regularized continuous fields by mapping pixel coordinates to output values [4]

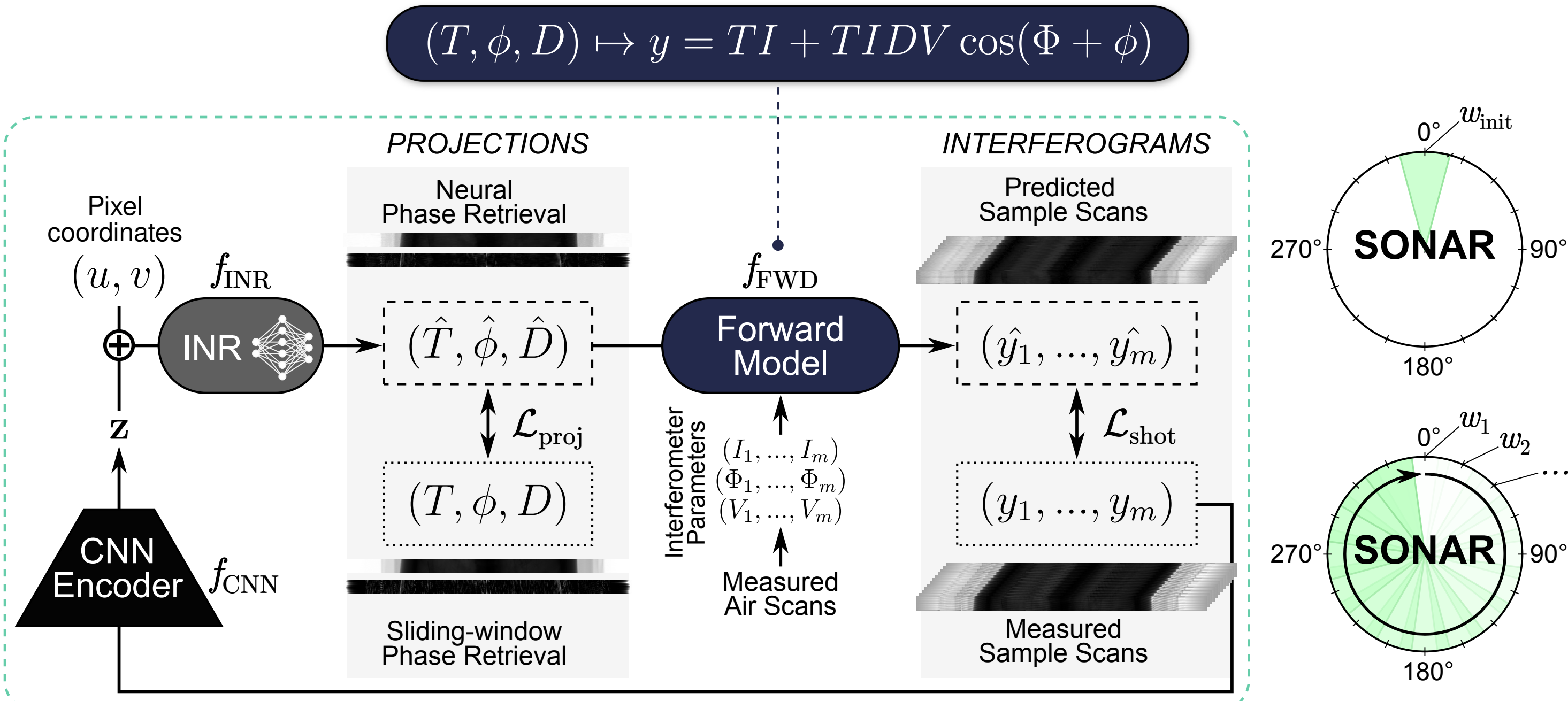


THE PHYSICS: TALBOT-LAU INTERFEROMETRY

- X-rays are diffracted through gratings to detect **wave-optical information: transmission T , phase shift ϕ , and dark-field D**
- Phase retrieval:** Ill-posed inverse problem requires multiple shots per angle, which however cannot be retrieved in a rotating gantry setup
- Conventional solution:** For each shot, adjacent scans are treated as if they were taken from the same angle (sliding-window phase retrieval)

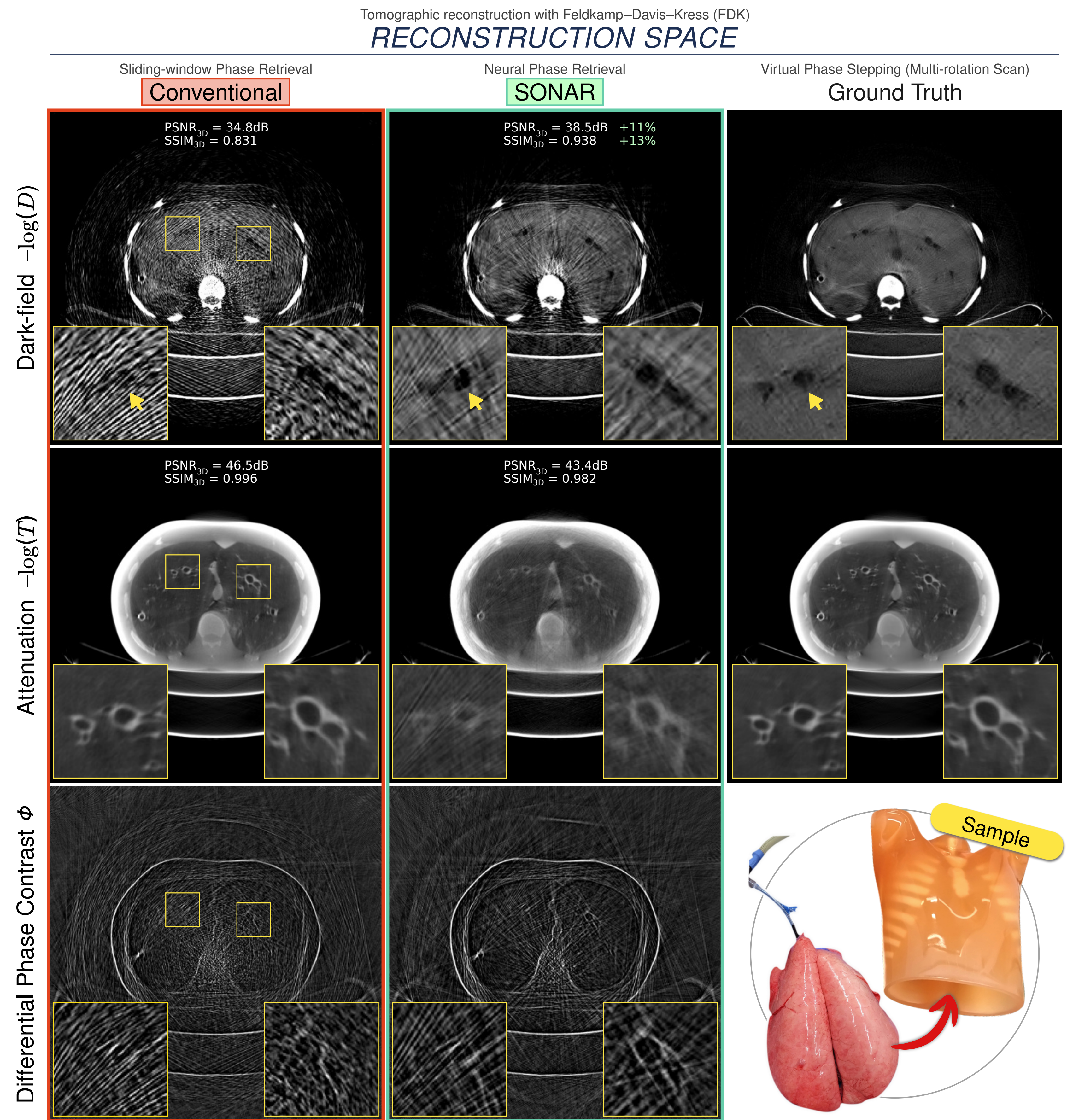
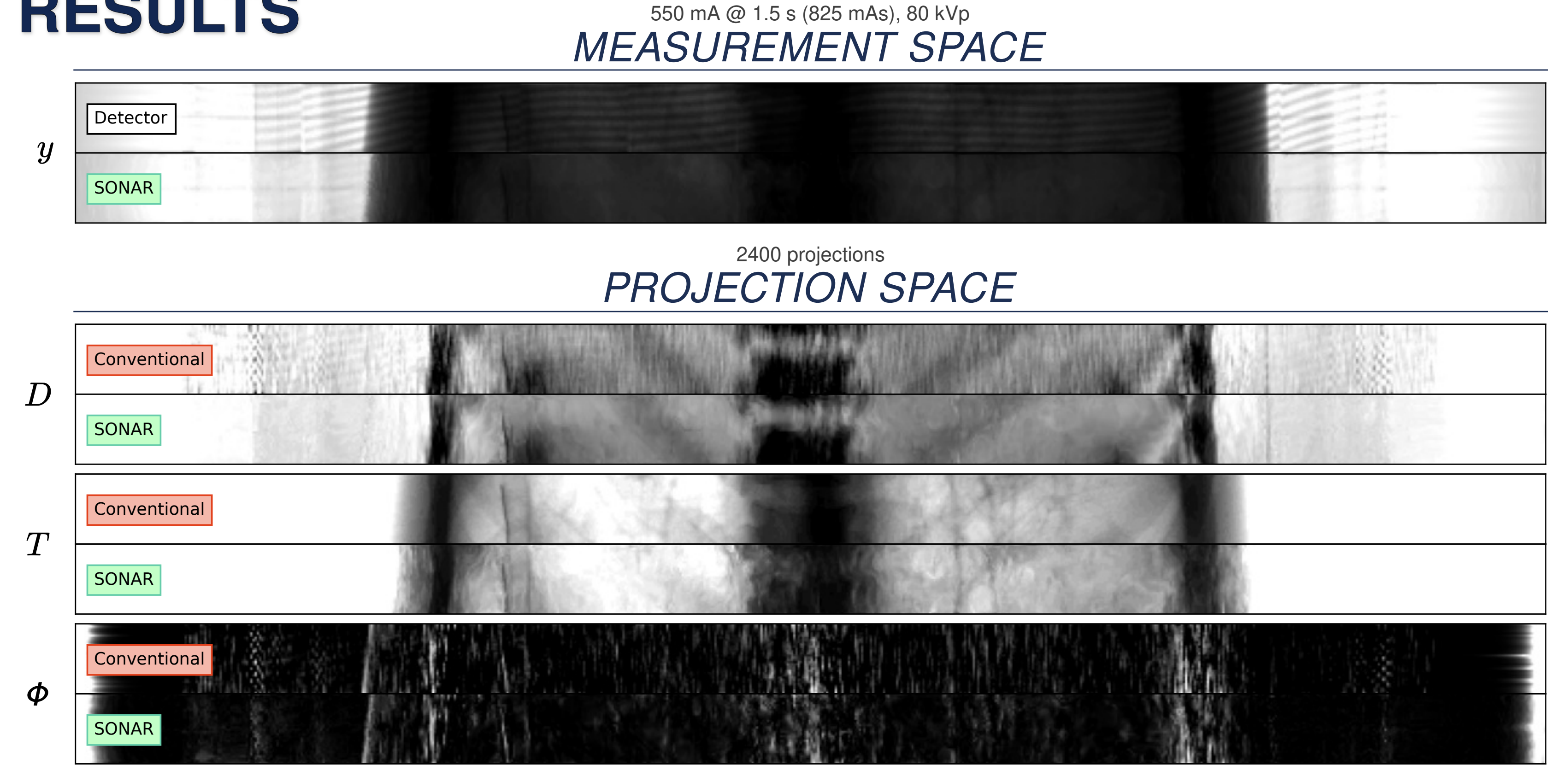


THE REPRESENTATION: MULTIMODAL NEURAL FIELD



- Physics-constrained** inverse problem solver with forward model, weak supervision based on the conventional phase retrieval results, and CNN encoder for spatial image coherence
- Training is initialized on a single shot (1000 iterations) followed by iterative **warm-start fine-tuning** (50 iterations) across all 2400 shots

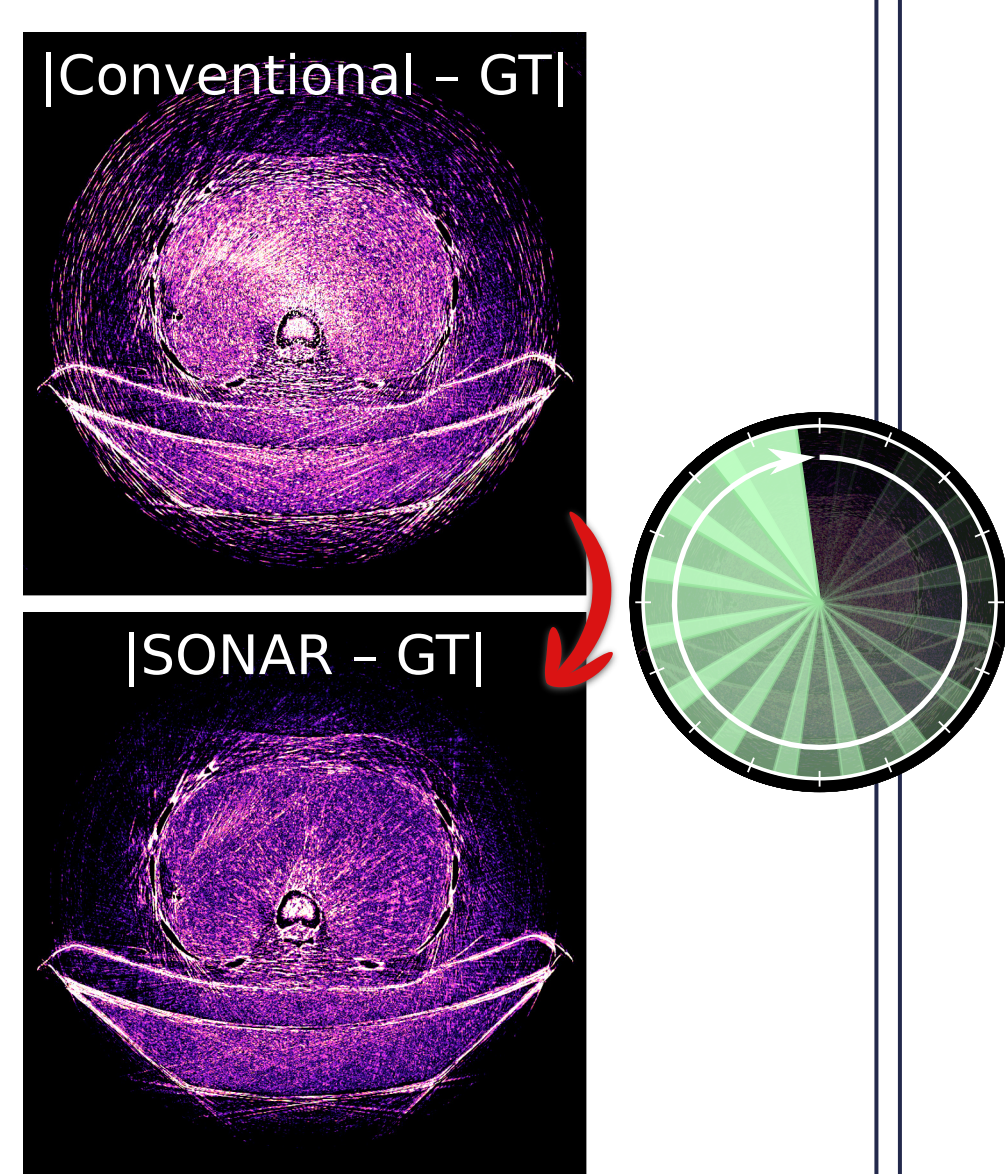
RESULTS



- SONAR enables **enhanced dark-field (and phase contrast) reconstruction quality**
- Multimodal neural field remains **physically consistent with detector measurements**

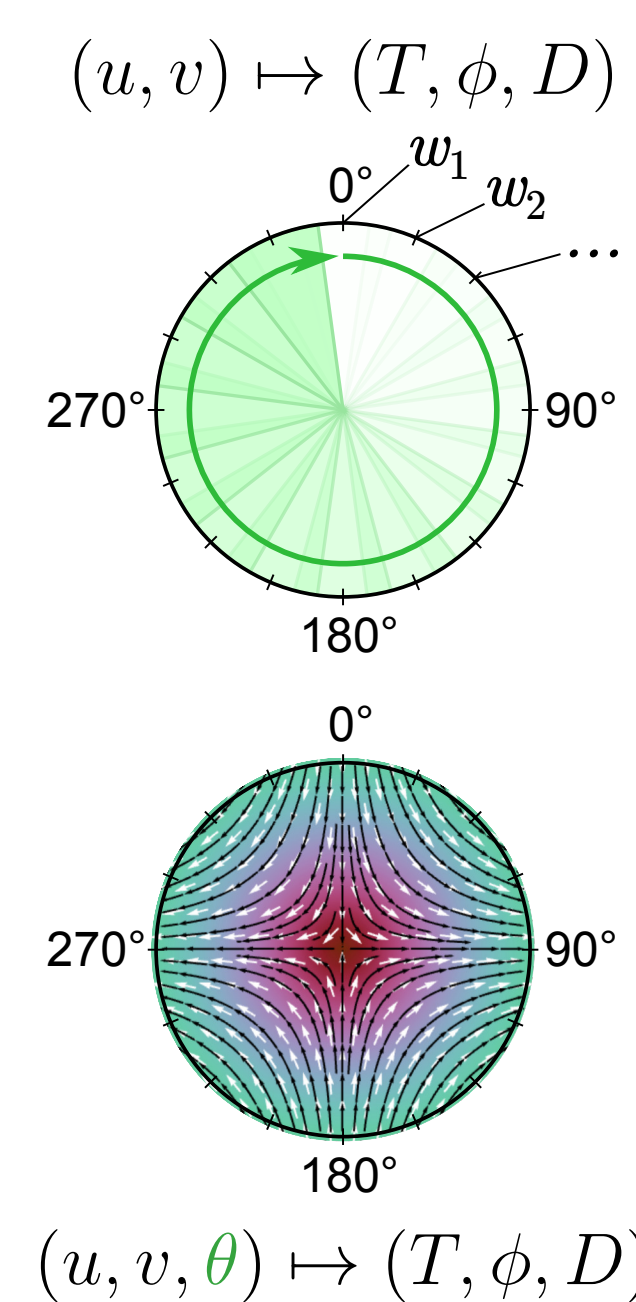
CONCLUSION

- We present the first physics-guided neural phase retrieval model for **feasible DFCT image quality**
- Self-supervised and per-instance** training mitigating the risk of hallucinations and omitting the need for ground truth
- Operation in the projection domain promotes **flexible frameworks** building explicitly on classical phase retrieval methods



OUTLOOK

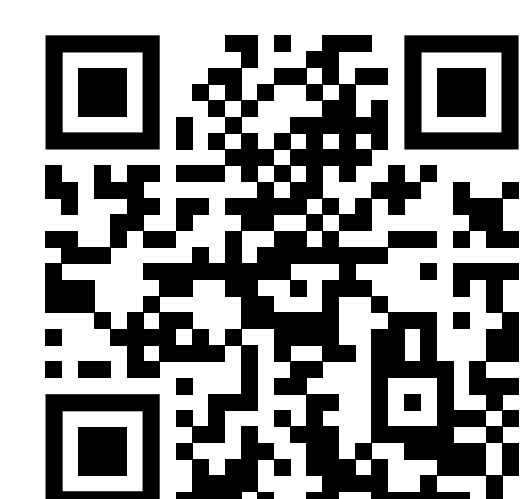
- Encoding angular dimension** directly into the INR to facilitate training
- Generalization** of hyperparameters across subjects and meta-learning
- Exploration of **spectral bias** modulation for regularization
- Application to other phase-sensitive X-ray imaging setups



REFERENCES

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Project webpage:
[dcfrey.github.io/sonar](https://github.com/dcfrey/sonar)
Corresponding author:
daniel.frey@tum.de
github.com/dcfrey